

BALAJI PERNI

(+91) 9182972833 | pernibalaji35@gmail.com | [LinkedIn](#) | [GitHub](#)

CAREER OBJECTIVE

Results-driven Data Science undergraduate specializing in **Computer Vision** and **Deep Learning** with proven experience building and deploying production-grade AI models. Adept at developing end-to-end ML pipelines — from data preprocessing to REST API deployment. Seeking to apply expertise in object detection, transfer learning, and model optimization to solve real-world problems at scale.

EDUCATION

SRKR Engineering College, Bhimavaram

2023 – 2027

Bachelor of Technology in Artificial Intelligence and Data Science

CGPA: 8.8 / 10

PROJECTS

Shrimp Count Prediction — *Government-Sponsored Project*

[\[Live Demo\]](#)

Apr – Aug 2024

- Trained **YOLOv8** object detection model on 8,000+ labeled images, achieving **91% mAP** at 24 FPS for accurate real-time shrimp counting in aquaculture pond images
- Built end-to-end inference pipeline using **OpenCV** for image preprocessing, detection, and automated count output with visual bounding-box overlays
- Tech Stack:** Python, YOLOv8, OpenCV, TensorFlow, Pandas, NumPy

Deepfake / AI-Generated Image Detection System

2024

- Built a **binary image classification** model using **PyTorch** and **ResNet50** transfer learning to accurately distinguish real photographs from AI-generated synthetic images
- Designed a structured end-to-end ML pipeline encompassing data preprocessing, augmentation, model training, validation, testing, and deployment-ready inference modules
- Applied fine-tuning, dropout regularization, and learning rate scheduling to reduce overfitting; consistently monitored accuracy and validation loss across training epochs
- Structured a modular, production-ready codebase with clearly separated components for training, evaluation, and model serving
- Tech Stack:** Python, PyTorch, ResNet50, OpenCV, NumPy, Scikit-learn

AI-Powered Task Management Platform

2024

- Designed and built a full-stack **REST API** using **Node.js/Express** with **PostgreSQL** and **Prisma ORM**, implementing JWT auth, role-based access control, audit logging, and hierarchical task/subtask workflow with automatic parent-child status sync
- Engineered a **FastAPI** ML microservice integrating **sentence-transformer** embeddings with a custom-trained **logistic regression** intent classifier to auto-generate intelligent subtasks, with confidence-based fallback to prevent hallucinations
- Applied production-level security practices including rate limiting, Helmet middleware, CORS configuration, environment variable isolation, and structured error handling across all service layers
- Tech Stack:** Python, Node.js, Express, PostgreSQL, Prisma, FastAPI, JWT, Scikit-learn, Docker

TECHNICAL SKILLS

Programming	Python, SQL, Java, C
ML / DL	TensorFlow, Keras, PyTorch, Scikit-learn
Computer Vision	OpenCV, YOLOv8, ResNet50, MobileNetV2, CNN Architectures
ML Techniques	Transfer Learning, Fine-Tuning, Object Detection, Image Classification, Model Deployment
Data & Tools	Pandas, NumPy, Git, Jupyter, VS Code, Google Colab
Databases	MySQL, MongoDB, PostgreSQL

CERTIFICATIONS & ACHIEVEMENTS

- Introduction to Generative AI** — *Google Cloud*
- Machine Learning A–Z** — *Udemy*
- Data Science Specialization** — *Udemy*
- SQL Certification** — *GeeksforGeeks*